
Supplementary information

**Genome-wide CRISPR screen in human
T cells reveals regulators of FOXP3**

In the format provided by the
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Additional and detailed Methods:

Analysis of RNA-seq data

RNA-seq sequencing and differential expression analysis

Raw fastq files were aligned to the human genome (GRCh38; GenBank assembly GCA_000001405.40) using the STAR¹ pipeline (v2.7.10b). HTSeq-count² (v2.0.2) was used to generate gene counts and genes were filtered to retain those with raw read counts ≥ 10 in at least 2 samples. The resulting gene expression matrix was subsequently normalized using the trimmed mean of M value (TMM) method.

Genes with low read counts were removed and DEG analysis was performed using the DESeq2³ (v1.42.1) package while accounting for covariates such as donor effects. The resulting *p*-values were adjusted for multiple hypothesis testing and filtered to retain DEGs with a *q*-value < 0.05 .

For GSEA analysis (**Fig. 3f**), we used the LFC values obtained from DESeq2 as input for enrichment with the fgsea (v1.26.0)⁴ package. Annotated positive or negative DEGs were derived from comparisons between in vitro expanded nTreg and Tconv and were used as the reference gene annotation database.

Processing and visualization of genomic data

Initial quality-control assessment was performed using the FastQC toolkit (v0.11.8; <https://github.com/s-andrews/FastQC>). For ATAC-seq and inATAC-seq, raw fastq files were first trimmed using Trim Galore (v0.6.4)⁵ to remove contaminating Illumina adapter sequences. Fastq files were aligned to the human genome (GRCh38; GenBank assembly GCA_000001405.40) using bowtie2 (v2.5.0)⁶ and processed with samtools (v1.17)⁷. Genomic tracks for ChIP-seq, ATAC-seq, inChIP-seq and inATAC-seq data were generated using the deepTools (v2.0)⁸ bamCoverage function and visualized on the WashU Epigenome Browser⁹. Peaks were called using macs2 (v2.2.9.1)¹⁰. CUT&RUN data was processed using the CUT&RUNTools (v2.0) pipeline¹¹, following recommended settings.

Analysis of Perturb-Seq data

Preprocessing

For the scRNA data, sequencing reads for all libraries were aligned to the hg38 reference genomes using cellranger (v7.0.0). Surface protein tag abundances and CROP-seq gRNA reads were estimated using the kallisto (v0.44.0), bustools (v0.39.3) and kite (v0.02) frameworks^{12,13}. Intracellular protein tag abundances were estimated using cellranger (v7.0.0).

Data processing

Data were analyzed using Seurat 4.1.2¹⁴. Low-quality cells were filtered based on standard quality-control metrics. In particular, cells were filtered based on mitochondrial content (allowing for a maximum of 5%), complexity (total RNA UMIs less than 3,000 or more than 50,000) and by the

number of detected genes (removing cells expressing less than 1,000 detected genes). Mitochondrial and ribosomal genes were excluded from subsequent analyses. RNA counts were normalized by total counts and log-transformed.

From the hashtag count matrix, we assigned donor identity using the *HTODemux()* function from Seurat (with *quantile = 0.90*) Cells with less than 50 total UMI counts from intracellular protein staining were discarded. All Protein assays were normalized with the centered log-ratio (CLR) method.

Guide RNAs were assigned by the following method¹⁵: gRNA identities with an UMI count greater than 4*SD (4*standard deviation) were assigned to each cell. Cells resulting with more than one assigned gRNA were considered as multiplets, while cells with less than 10 UMI counts were considered negative.

Regulatory coefficient inference

We used the MIMOSCA¹⁶ and Perturb-CITE-seq¹⁷ frameworks to infer the regulatory effect of each gRNA on the transcriptome. Briefly, this analysis includes two major iterative steps. The first step is a linear regression (LR), via an elastic net regularization, which generates a regulatory coefficient matrix. In addition to gRNA assignment, the LR accounts for both technical and biological variations through the evaluation of confounding factors, namely the total number of UMI counts mitochondrial content, donor batch and cell cycle phase (previously calculated with *CellCycleScoring()* from Seurat). The second step infers the probability of perturbation for each cell through an expectation maximization (EM)-like procedure.

The first LR was performed on 3,074 features, selected in accordance with a previous study¹⁸. A relative *z*-normalization towards the non-target cells (NTC) was applied to each library. Then, an Anderson-Darling test was performed (with the *kSamples* v1.2 R package) for each gRNA, and asymptotic *p*-values were corrected with Benjamini-Hochberg (BH) method (with the *multtest* v3.20 R package¹⁹). The union across all the different gRNAs of the respective top 10 genes with an adjusted *p*-value less than 0.1 was finally selected for LR.

Following the first LR, perturbation probabilities were inferred, and the obtained covariate matrix was submitted for a second LR. At this step, a correlation matrix towards gRNAs was computed on the regulatory coefficient matrix. Correlations of gRNAs targeting different genes were used to build a null distribution and ultimately calculate the empirical cumulative distribution function and empirical *p*-values for each pair of gRNAs targeting the same gene. All the gRNAs belonging to at least one gRNA pair with a BH adjusted *p*-value less than 0.05 were kept for the next LR, which performed on the 228 selected targets (KOs). As previously, perturbation probabilities were calculated once again, leading to a final LR step. For further analyses, only cells with a perturbation

probability over 0.5 were kept. To associate KO targets with FOXP3 protein, LR was performed on the protein FOXP3 as well. An alpha coefficient of 0.0001 was used for all LR.

For each target, regulatory coefficients obtained after permutations ($n = 100$) of cell assignment (between KOs and NTC) enabled to build a coefficient null distribution and thus calculate empirical p -values.

Module definition

Targets displaying only null coefficients were discarded. Additionally, genes with less than 4 absolute coefficients greater than 0.08 were discarded as well. Both target and gene modules were defined based on k means clustering ($k = 20$, 500 initializations). Clusters with similar coefficient profiles were merged, defining in total 11 and 10 target and gene clusters, respectively. UMAP coordinates were calculated with *umap*. Graphs and centrality measures were generated and calculated with *igraph 1.4.2*.

Protein expression score

For each target i and each protein, the expression score was calculated as follow:

$$Protein\ score(i) = \frac{mean(protein)_i - mean(protein)_{NTC}}{mean(protein)_{NTC}}$$

Differential expression analysis

DGE analysis was performed on RNA and protein assays, using the Seurat function *FindMarkers()*. For DGE on FOXP3⁺ cells, single cell transcriptomes were enriched on the basis of FOXP3 protein expression. The top 40%-expressing cells for a given genetic perturbation or control cells were taken for downstream analyses. Other percentiles were tested to ensure the robustness of our method.

Gene score

Gene signature scores were calculated with the Seurat function *AddModuleScore()* and compared using a Wilcoxon test. Cells were sorted on FOXP3 protein expression as described above. The human nTreg vs Tconv gene signature was defined above.

Comparison between icRNA-seq and Perturb-icCITE-seq

LogFC of DEGs between RBPJ-KO and NTC in bulk icRNA-seq ($|\log_2(FC)| \geq \log_2(1.25)$ and adjusted p value ≤ 0.01) were compared with ones obtained from the Perturb-icCITE-seq data (FOXP3⁺ cells, see above). Lowly-expressed genes (i.e. detected in less than 10% of the cells for both conditions in the single-cell data) were filtered out for this analysis.

In-house icCITE-seq antibody conjugation

TotalSeq-style oYo-Link® ssDNA oligonucleotides (AlphaThera) were ordered with sequences of:

GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTNNNNNNNNNNNNXXXXXXXXXXXXX
XXNNNNNNNNNGCTTTAAGGCCGGTCCTAGC*A*A for antibody tags where X's represent the antibody barcode and N's represent UMI tags. ssDNA oligonucleotides were covalently linked to purified antibodies by UV as per the manufacturer's instructions. Excess oligos were removed from conjugates using a centrifugal filter column (MilliporeSigma; UFC30HV00) with an IgG Sepharose slurry (GE Healthcare; 17096902) as per manufacturer's recommendations. Each conjugation was confirmed through running ~1 µg of conjugated antibody on a 4-12% protein gel (Thermo; NP0321BOX) and staining with Imperial™ Protein Stain (Thermo; 24615).

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